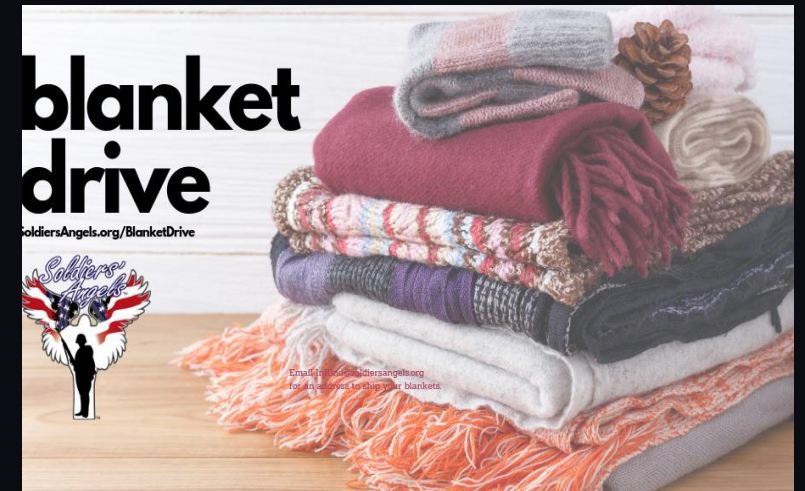


Bear or Blanket? (Automated Donation Sorting)



Team Members:

Kala Hayes, Zaha Muhammad, Andy Nguyen,
Tirimisiyu Tewogbola, Yolanda Armstrong

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Objective: To leverage Computer Vision to speed up the delivery of essential bedding to veterans transitioning into housing.

Scenario: A sorting assistant for veterans charities (like Project Jumpstart) to separate "Comfort Item" from "Essential Bedding."

The Problem & Solution

The Problem: Donation centers receive massive volumes of unsorted "soft goods" (bedding vs toys).Manual sorting task ~2 hours per bin, delaying veterans from receiving basic sleep essentials on their first night in housing.

Our Solution: A classification system that distinguishes between Teddy Bears (Comfort) and Blankets (Essentials).

CV Distinguishing :distinguishes between Teddy Bears (Comfort)and Blankets (Essentials) when both share fuzzy, and softtextures.

Impact: Automating the first stage of sorting allows volunteers to focus on distribution, saving time, and improving delivery speed

Dataset Analysis & Fixes

Data Profile: 349 total custom images

- Blankets: 237 (68% of data)
- Teddy Bears: 112 (32% of data)
- The Challenge: We identified a Class Imbalance (2:1 ratio) that risked model Bias.
- The Fixes:
 - Data Augmentation: Used random rotations, zooms, and crops to "multiply" the minority bear class.
 - Class Weighting: Implemented `class_weight= {0: 1.0, 1: 2.1}` to force the model to pay equal attention to the smaller bear dataset.

Technical Approach

- Technique: Binary Image Classification
- Model: ResNET50 (via Transfer Learning)
- Framework: TensorFlow /Keras (Google Colab T4 GPU)
- Justification: We used a pre-trained "Expert" model because it already understands textures (LBP) and shapes (HOG). This allowed us to achieve results with a smaller, custom dataset of 349 images rather than needing thousands.

Validating & Tuning

Training: Conducted 15 epochs using data augmentation (rotating and zooms) to prevent the model from just memorizing the photos.

Validation Results: Achieved a 67.6% Validation Accuracy. This is realistic baseline for custom "real-world" data where objects look very similar.

5/10/2026

Final Results

```
64
65 x 7. VISUALIZE RESULTS
66 plt.figure(figsize=(12, 4))
67 plt.subplot(1, 2, 1)
68 plt.plot(history.history['accuracy'], label='Train Accuracy')
69 plt.plot(history.history['val_accuracy'], label='Val Accuracy')
70 plt.title('Accuracy: Teddy Bear vs Blanket')
71 plt.legend()
72
73 plt.subplot(1, 2, 2)
74 plt.plot(history.history['loss'], label='Train Loss')
75 plt.plot(history.history['val_loss'], label='Val Loss')
76 plt.title('Loss: Donation Sorting Precision')
77 plt.legend()
78 plt.show()
79
```

Found 276 images belonging to 3 classes.
Found 71 images belonging to 3 classes.
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop_94/962746/962746.h5
Starting to train... 16 Run/step

Epoch	12h/step	accuracy	loss	val_accuracy	val_loss
Epoch 1/15	12h/step	0.3813	1.1387	0.6761	0.6572
Epoch 2/15	12h/step	0.5548	0.9391	0.6761	0.6239
Epoch 3/15	13h/step	0.6439	0.8528	0.6761	0.6238
Epoch 4/15	14h/step	0.5576	0.9832	0.6761	0.6358
Epoch 5/15	14h/step	0.5396	0.9959	0.6761	0.6377
Epoch 6/15	14h/step	0.5468	0.8983	0.6761	0.6226
Epoch 7/15	12h/step	0.6187	0.8438	0.6761	0.6263
Epoch 8/15	13h/step	0.5895	0.8381	0.6761	0.6288
Epoch 9/15	14h/step	0.5827	0.8386	0.6761	0.6312
Epoch 10/15	13h/step	0.5791	0.8424	0.6761	0.6263
Epoch 11/15	14h/step	0.5252	0.8691	0.6761	0.6376
Epoch 12/15	14h/step	0.5827	0.8863	0.6761	0.6489
Epoch 13/15	13h/step	0.5334	0.8381	0.6761	0.6481
Epoch 14/15	13h/step	0.5863	0.8449	0.6761	0.6263
Epoch 15/15	32h/step	0.5882	0.8257		

- The "Donation-bot" achieved a 67.6% validation accuracy, successfully automating identification of high-priority bedding for veteran move-in kit.
- By handling over two-thirds of the sorting automatically, this tool significantly reduced volunteer labor and speed up supply delivery to homeless veterans.
- Failure Analysis: The "white bear" case
 - The model struggled with Texture Overlap (LBP); the long white fur of the teddy bear mathematically identical to white fleece blankets
 - When bears were "scrunched" or viewed from the back, they lacked Semantic Features (eyes/nose), causing the AI to default to the majority class: Blanket.

